

Ex73_NoiseCancelling

April 19, 2026

1 MSE StatDig : Chap 7 “Optimum filter”

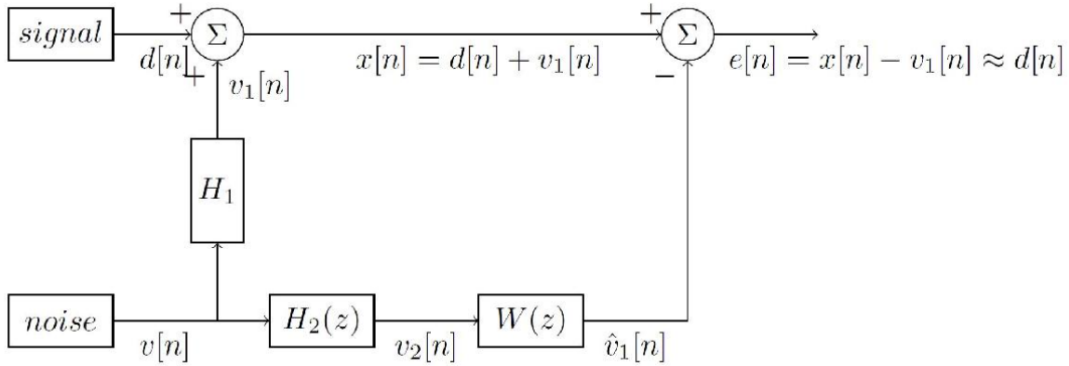
1.1 Ex 7.3 FIR Wiener Noise Cancelling

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1.2 General

1.2.1 Description

The noise cancellation problem is described by the following figure:



Suppose the desired signal $d[n]$ is given by :

$$d[n] = \sin(n\omega_0 + \phi)$$

with $\omega_0 = 0.05\pi$ and the related noise sequence $v_1[n]$ and $v_2[n]$ are AR(1) processes driven by the same noise signal $v[n]$.

$$v_1[n] = 0.8 v_1[n-1] + v[n]$$

$$H_1(z) = \frac{1}{1 - 0.8z^{-1}}$$

$$v_2[n] = -0.6 v_2[n-1] + v[n]$$

$$H_2(z) = \frac{1}{1 + 0.6z^{-1}}$$

Where $v[n]$ is zero-mean, unit variance white noise that is clearly uncorrelated with $d[n]$. Goal is to design a FIR Wiener filter of orders 5 and 11 to remove the noise.

1.2.2 Work

Ex1 : Signal generation

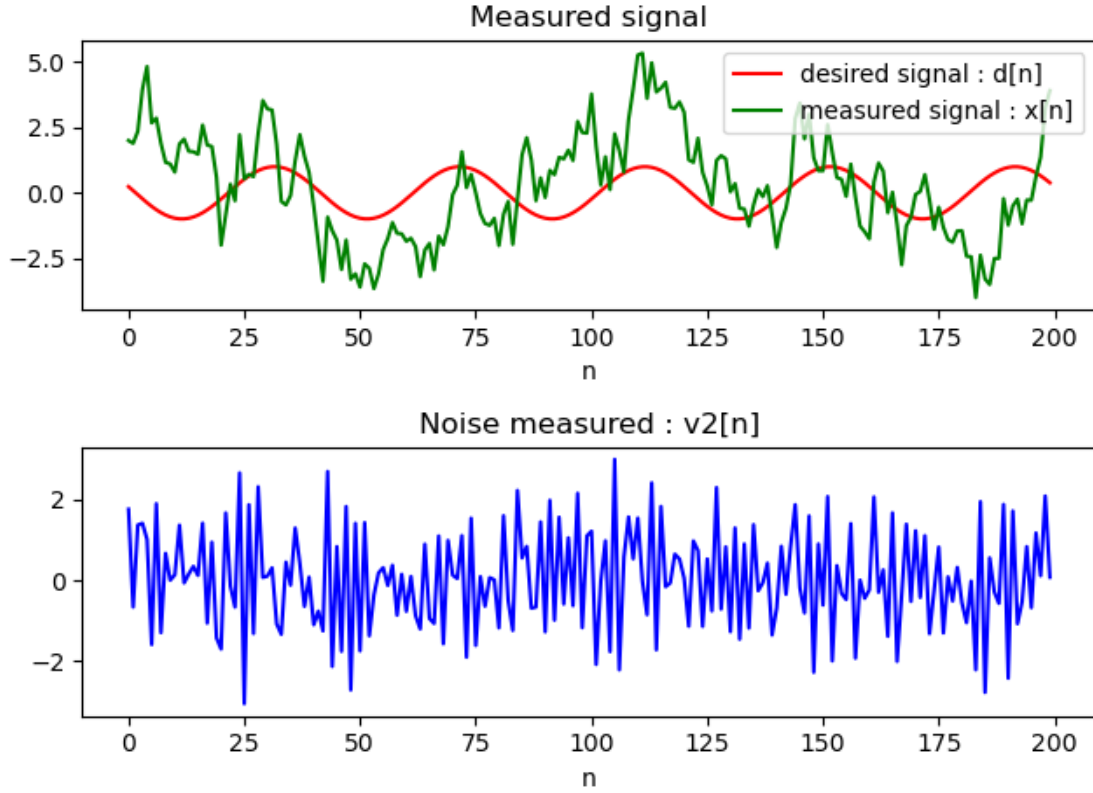
- Display the signals created by the following code :

```
worders = [5,11] # filter orders
N = 200 # number of sample
phi = np.random.rand()*2*np.pi-np.pi # phase of the signal
n = np.arange(N)
SigmaN = 1 # std of the noise
H1_a = -0.8 # AR(1) coeff of H1
H2_a = 0.6 # AR(1) coeff of H2

# fix the random seed for reproducibility
np.random.seed(0)

w0 = 0.05*np.pi;
d = np.sin(n*w0+phi); # Signal without noise
v = SigmaN * np.random.normal(0,1,N) # source noise

v1 = signal.lfilter(1,[1,H1_a],v) # noise which disturb d[n]
v2 = signal.lfilter(1,[1,H2_a],v) # noise measured
x = d+v1; # Measured signal
```



Ex2 : Wiener solving

Ex2.1 Filter design The Wiener-Hopf equations for the noise cancellation described by figure are given by :

$$\mathbf{R}_{v_2 v_2} \mathbf{w} = \mathbf{r}_{v_1 v_2}$$

As we assume that the noise ($v_2[n]$) is uncorrelated with ($d[n]$) :

$$r_{v_1 v_2}[k] = \mathbb{E} \{v_1[n]v_2^*[n-k]\} = \mathbb{E} \{(x[n] - d[n])v_2^*[n-k]\}$$

Then, the Wiener-Hopf equations become:

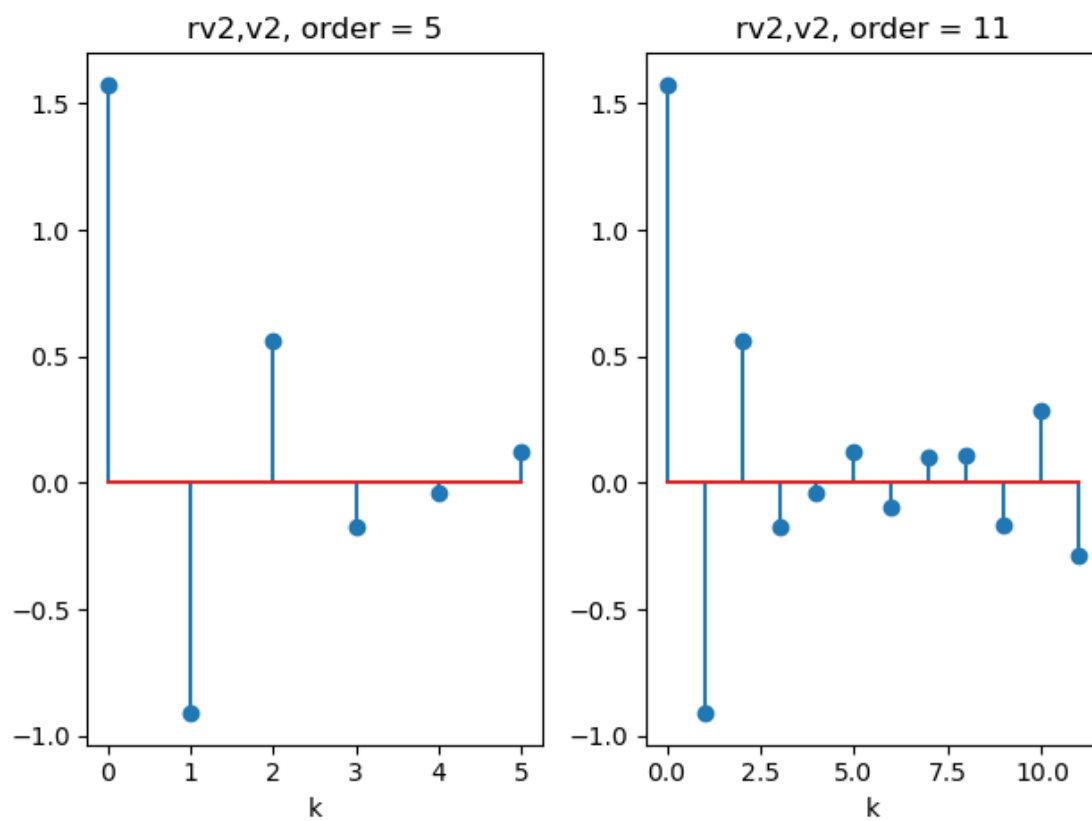
$$\mathbf{R}_{v_2 v_2} \mathbf{w} = \mathbf{r}_{x v_2}$$

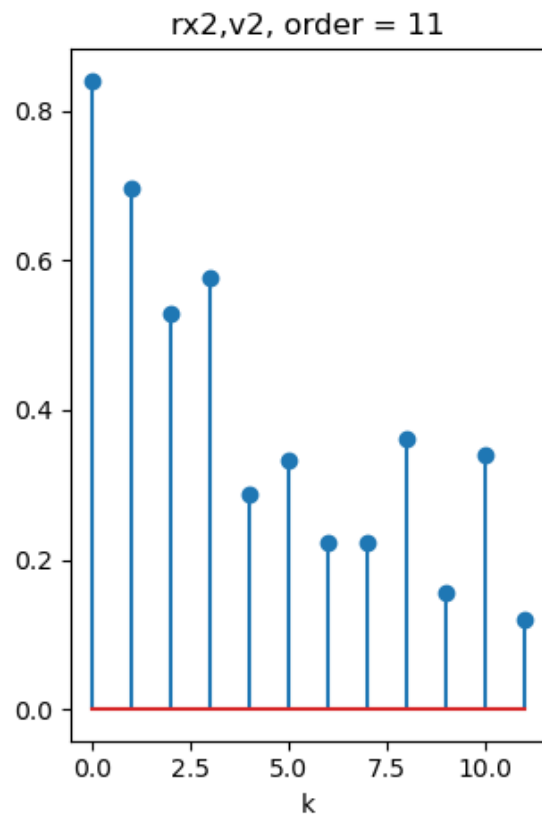
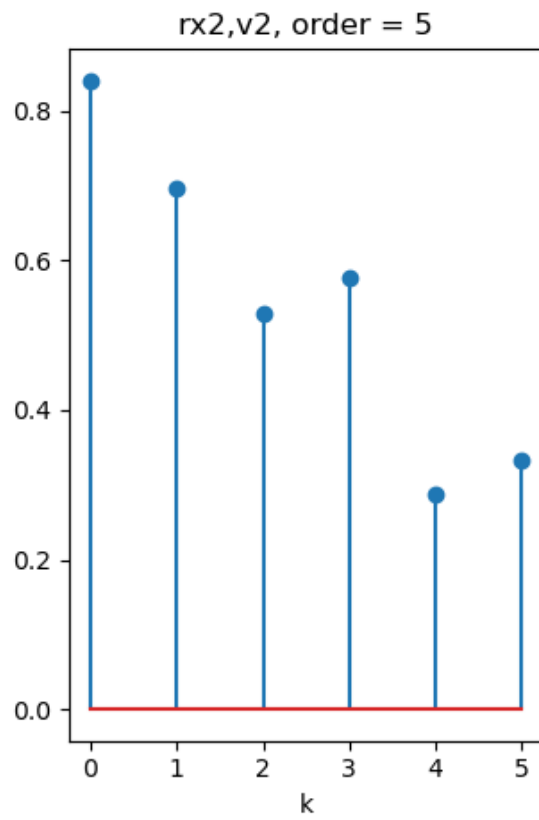
And the solution is:

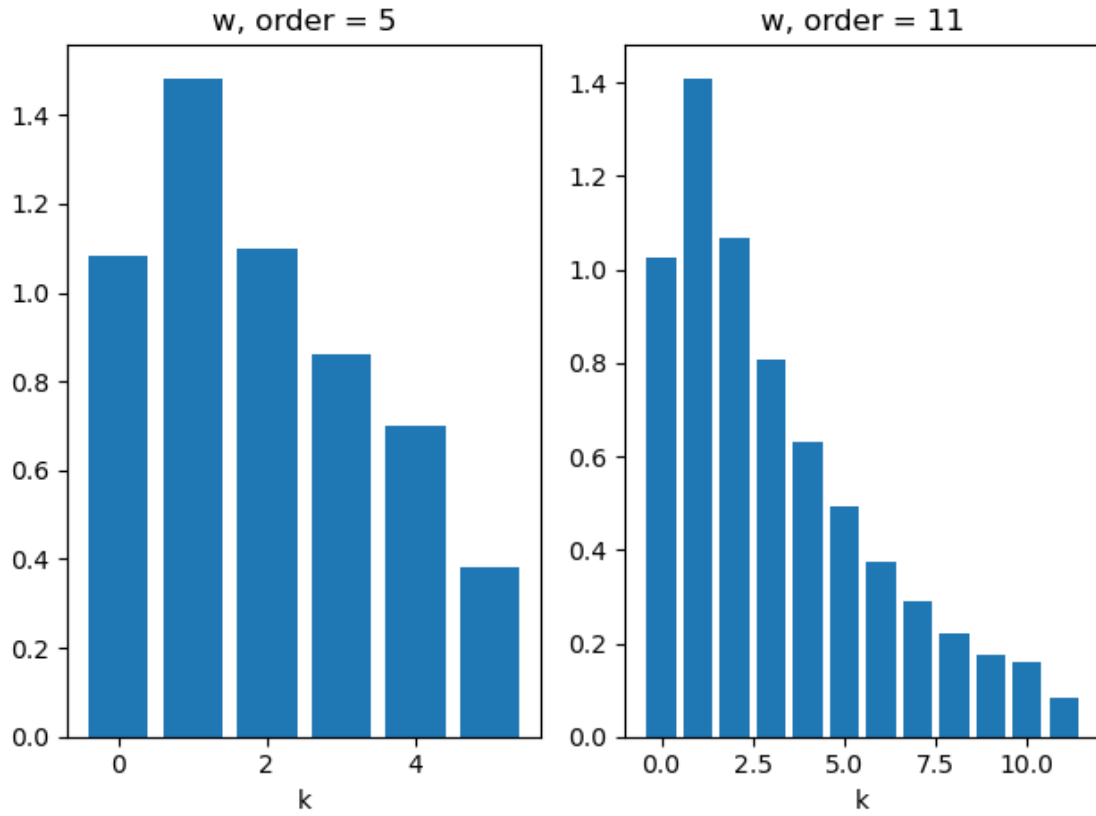
$$\mathbf{w} = \mathbf{R}_{v_2 v_2}^{-1} \mathbf{r}_{x v_2}$$

- Compute and display the correlation vectors r_{v_2, v_2} and r_{x, v_2} for each filter order

- Compute and display the filter coefficients for each order



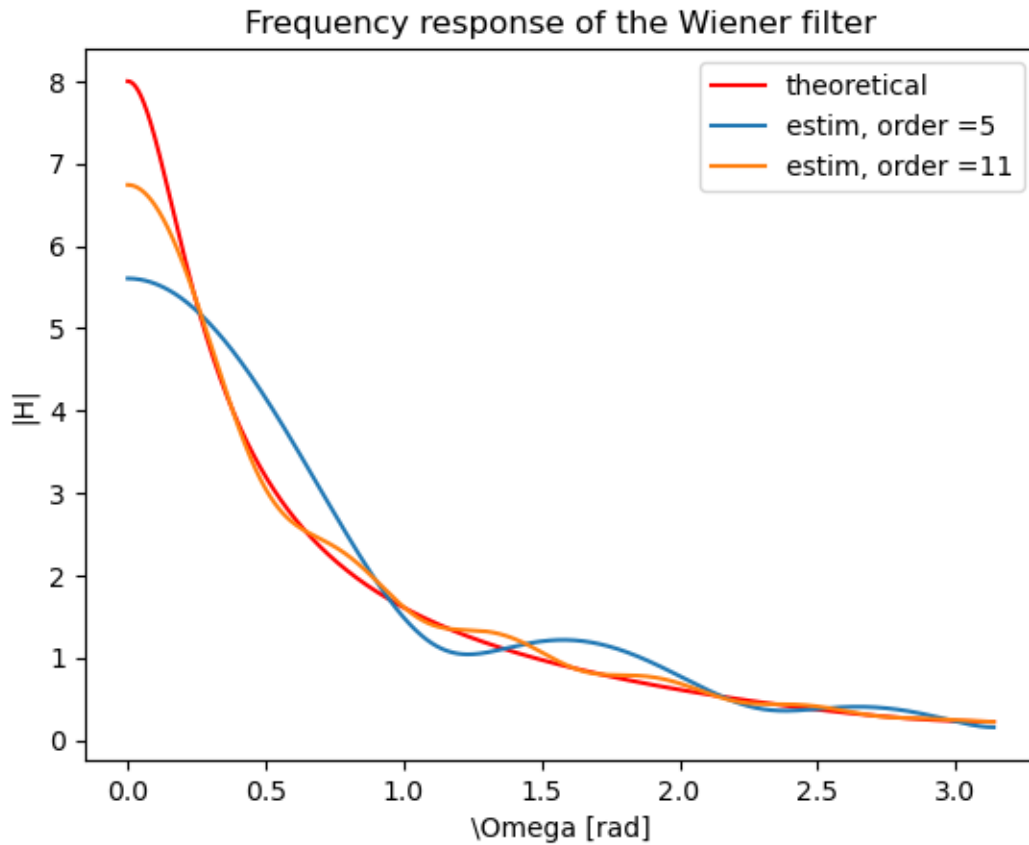




Ex2.2 Frequency response

- compute and display the frequency response for each filters
- compute and display the theoretical frequency response $H(f)$

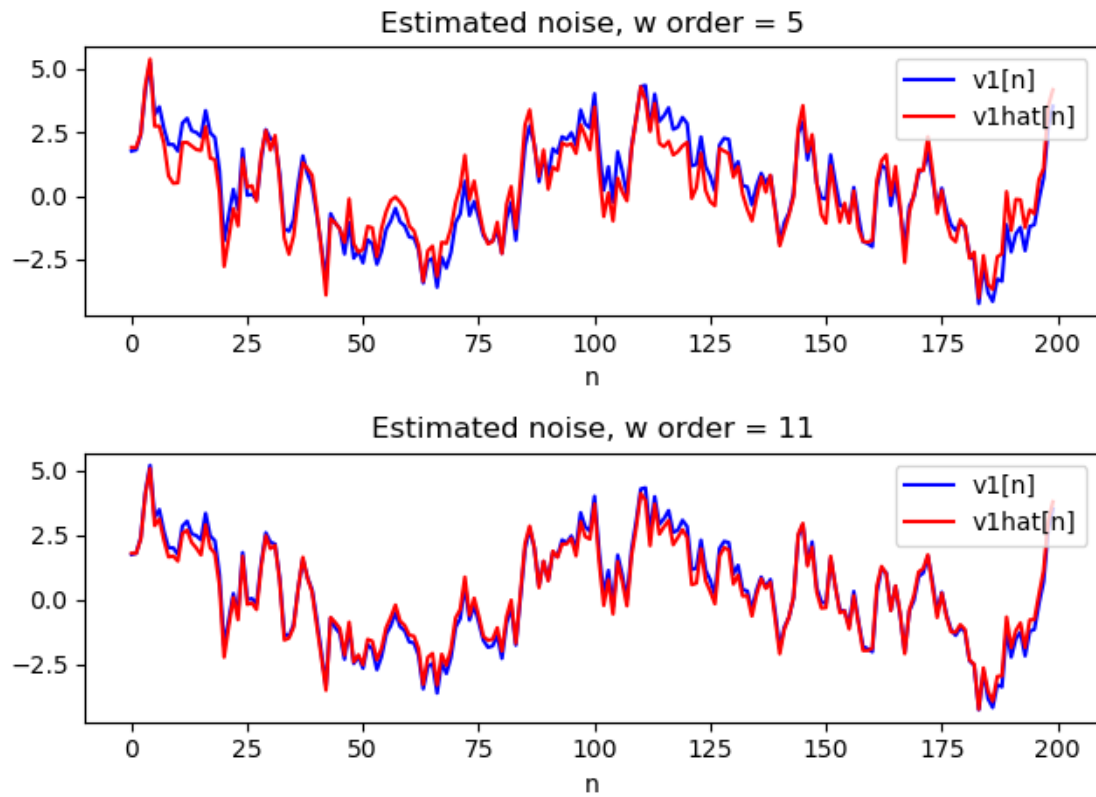
$$H(f) = \frac{H_1(f)}{H_2(f)}$$



Ex2.3 filtering

- Estimate and display the noise $v_1[n]$ based on $v_2[n]$ and the Wiener filter designed

```
[1.08138211 1.48294386 1.09811131 0.8599835 0.70056758 0.38208944]
[1.02575807 1.40951806 1.06759939 0.80656862 0.62969432 0.49270844]
0.37297802 0.29041823 0.22296058 0.17693976 0.16169111 0.08422758]
```



Ex2.4 Signal reconstruction Finally, the estimated noise can be removed from the measured signal.

- Compute and display the reconstructed signal
- Compute and display the approximation error

